Normal operation, after use of a fallback arrangement, should only be restored upon confirmation by the operator.

12.3 Failure or change of sensor

12.3.1 The failure or change of a sensor shoul d not result in sudde n changes of control

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SOURCE: MSC.252(83).pdf
Normal operation, after use of a fallback arrangement, should only be restored upon confirmation by the operator.

12.3 Failure or change of sensor

12.3.1 The failure or change of a sensor shoul d not result in sudde n changes of control commands or loss of manoeuvring control. This may be accomplished by appropriate integrity checks using the information from several sources.

12.3.2 In case of a sensor or source failure, the sy stem should provide an al ert and indicate (an) alternative sensor(s) or source(s), as available.

12.3.3 If sensors or sources are not able to provide nece ssary ship status or navigation data for automatic control functions, a dead reckoning procedure should provide the missing information, as far as practicable.

12.4 Storage of system related parameters

All system related parameters and settings should be stored in a protected way for reconfiguration of the INS.

12.5 The automatic response to malfunctions should result in the safest possible configuration accompanied by an alert.

12.6 Alert management

12.6.1 System failures should be alerted according to the requirements described in Module C.

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SOURCE: MSC.252(83).pdf
12.5 The automatic response to malfunctions should result in the safest possible configuration accompanied by an alert.

12.6 Alert management

12.6.1 System failures should be alerted according to the requirements described in Module C.

12.6.2 Loss of system communication between th e alert management and the navigational systems and sensors should be indicated as a warning at the central alert management HMI.

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12.6.3 A system failure of the alert management or the loss of system communication between the alert management and the navigational functions , sources and/or sensor s, should not lead to the loss of the alert announcement functionali ty of the individual navigational functions, sources/sensors.

12.7 Fallback for navigational information failure

12.7.1 In the event of failures of navigational information and to maintain minimum basic operation,

there should be a permanent indication of the failed input information and the fall-

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SOURCE: MSC.252(83).pdf
12.7 Fallback for navigational information failure

12.7.1 In the event of failures of navigational information and to maintain minimum basic operation,

there should be a permanent indication of the failed input information and the fall-back activated,
the respective actions of the alert management should be activated, and
the fallback arrangements listed below should be provided.

12.7.2 Route monitoring

12.7.2.1 Failure of heading information (Azimuth Stabilization)

The INS should display own ship’s position and ove r-ground-motion vector in the chart and not the ship’s heading line.

12.7.2.2 Failure of course and speed over ground information

The INS should display own ship’s position and heading line.

12.7.3 Collision avoidance

In the case of failure of:

Heading information
Speed through the water information
Course and speed over ground information
Position input information
Radar video input information
AIS input information,

the INS should operate as defined in the op erational Module B4 of the proposed modular

structure for radar performance standards as set out in appendix 3.

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SOURCE: MSC.252(83).pdf
Radar video input information
AIS input information,

the INS should operate as defined in the op erational Module B4 of the proposed modular structure for radar performance standards as set out in appendix 3.

12.7.4 Heading/Track control

The requirements for the applicab le control function as specifie d in the individual performance standards should apply.

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13 Technical requirements

13.1 General

13.1.1 In addition to meeting the rele vant requirements of resolution A.694(17) , the INS should comply with the requirements of these performance standards.

13.1.2 Means should be provided to monitor and to display hardware malfunctions of the INS. Alerts should be provided in case of malfunctions.

13.2 Requirements for hardware and/or processors

13.2.1 Sensor

13.2.1.1 A sensor or part thereof is not part of the INS, if it only supplies raw data.

13.2.1.2 Processing of raw data from sensors may be part of the INS.

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SOURCE: MSC.252(83).pdf
13.2 Requirements for hardware and/or processors

13.2.1 Sensor

13.2.1.1 A sensor or part thereof is not part of the INS, if it only supplies raw data.

13.2.1.2 Processing of raw data from sensors may be part of the INS.

13.2.1.3 In case sources perform functions of th e INS these functions and interfaces should

conform with the relevant parts of these performance standards.

13.2.2 Actuator and controller

An actuator, controller or part thereof is not part of the INS, if it only receives data or commands and does not perform other functions of the INS as required by these standard.

13.3 Requirements for software

13.3.1 The operational software s hould fulfil the requirements of the relevant international standards related to maritime navigation and communication equipment.

13.4 Requirements for power supply

13.4.1 Power supply requirements applying to pa rts of the INS as a result of other IMO requirements should remain applicable.

13.4.2 The INS including the sensors for position, sp eed, heading and depth should be supplied:

.1 from both the main and the emergency source of electrical power with automated

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